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Riverscape genetics of the orangethroat darter complex

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Abstract

Freshwater darters belonging to the orangethroat darter species complex, or *Ceasia*, are widely distributed in the Central and Southern United States, with ranges that span both glaciated and unglaciated regions. Up to 15 species have been recognized in the complex, with one, *Etheostoma spectabile*, having a widespread northern distribution and another, *Etheostoma pulchellum*, having a sizeable southern distribution. The other species in the complex have much more restricted distributions in unglaciated regions of the Central Highlands. We sampled 384 darters from 52 sites covering much of the range of *Ceasia* and evaluated patterns of genetic diversity, genetic structure, and pre- and post-glacial patterns of range contraction and expansion. We anticipated finding much stronger signals of genetic differentiation and diversification in unglaciated regions, given the higher species diversity and levels of endemism reported there. Surprisingly, microsatellite genotyping revealed two well-differentiated genetic clusters of *E. spectabile* in samples from glaciated regions, one confined to the Illinois River basin and another found in the Wabash drainage and Great Lakes tributaries. This suggests that there was expansion from two isolated glacial refugia, with little subsequent post-glacial gene flow. Fish collected from throughout the unglaciated region were less genetically differentiated. Fish assigned to *Etheostoma burri* and *Etheostoma uniporum* based on collection sites and morphological characters were not genetically differentiated from *E. spectabile* samples from the region. Hybridization and introgression occurring in the Central Highlands may confound genetic delineation of species in this region of high endemism and diversity.

KEYWORDS
AIC, biogeography, *Ceasia*, cluster analysis, DNA microsatellites, glacial refugia

1 | INTRODUCTION

In North America, Pleistocene glaciations occurring between 2.6 million and 10,000 years ago had a lasting impact on many freshwater fish species. The cycles of glacial advance and retreat influenced habitat connectivity and configuration by altering drainage patterns and disrupting species' dispersal routes, migratory patterns, and life history (Larson & Schaetzl, 2001). The ranges of northern fish species contracted into various glacial refugia, and historic gene flow patterns between populations were different from those of contemporary patterns. After the Wisconsin glacial retreat (starting 15–12,000 years ago), populations expanded northward into new streams. In addition, unglaciated streams were periodically shifting and undergoing events such as stream capture, so southern populations were also dynamic

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(Mayden, 1985, 1988). Riverscape genetics aims to characterize the genetic structure of freshwater populations, identify current and historical gene flow patterns, detect barriers to movement, and highlight factors shaping diversification in the riverine environment (Davis et al., 2018). Here we apply riverscape genetics to a North American freshwater fish clade, the orangethroat darter complex. We ask if the legacy of glacial history is still evident in the genetic structure of current populations, and whether species within the complex formed distinct genetic clusters.

Darters (Percidae: Etheostomatinae) are one of the most diverse and abundant groups of North America fish fauna with approximately 248 described species (*c*. 174 *Etheostoma* species), representing one-fifth of North American fish diversity (Stepien & Haponski, 2015). Darters are typically a few centimeters in length as adults, and the males of many species display vibrant colouration during the breeding season. Geographic range sizes vary dramatically, spanning multiple drainages or limited to small sections of a single watershed. Species may display regional or basin-specific variation, sometimes so substantial that multiple races, clades, subspecies, or even species can be derived from a single species. These derived groups typically become classified as a species group or complex (Lang & Mayden, 2007).

The *Oligocephalus* subgenus has at least four proposed species complexes, including the orangethroat darter complex, which is referred to as the clade-based name *Ceasia* (Near et al., 2011). The complex contains 15 allopatric species originally delineated by morphological characteristics (Black, 1940; Ceas & Burr, 2002; Ceas & Page, 1997; Distler, 1968; Linder, 1955; Wiseman et al., 1978). Only two species, *Etheostoma spectabile* (Agassiz, 1854) and *Etheostoma pulchellum* (Girard, 1859), have ranges that span multiple watersheds, and *E. spectabile* is also the only species in the complex that has a substantial portion of its range in areas that were glaciated during the last glacial maximum (LGM). The remaining species in *Ceasia*, aside from *E. pulchellum*, are limited to highly localized watersheds in the unglaciated Highlands, an area with some of the highest diversity of freshwater fishes in North America (Mayden, 1985; Near & Keck, 2005).

Behavioral isolation, especially by males, reportedly drives reproductive isolation and speciation in the *Ceasia* clade. Differences in male morphology and colouration led to the elevation of several subspecies to species (Ceas, 1997; Ceas & Burr, 2002; Ceas & Page, 1997), and recent research suggests, surprisingly, that male colouration functions more in male–male competition than in female choice (Moran et al., 2017; Moran & Fuller, 2018a, 2018b; Zhou & Fuller, 2016). Males of *Ceasia* species discriminate against sympatric heterospecific males, whereas females do not (Moran et al., 2017). Selection to avoid hybridization with *Etheostoma caeruleum* Storer, due to low survival of backcross (F2) individuals, likely drives these strong male mating preferences in *Ceasia* that co-occur with *E. caeruleum* (Moran et al., 2018; Moran & Fuller, 2018a). However, lack of discrimination by females likely explains hybridization with other *Etheostoma* in the wild and even with other darter genera in laboratory settings (Bossu & Near, 2009, 2013; Branson & Campbell, 1969; Distler, 1968; Echelle et al., 1974; George, 2018; Hubbs & Strawn, 1957; Moran et al., 2020; Ray et al., 2008). Hybridization has led to genetic introgression into *Ceasia*; notably the *E. caeruleum* mitochondrial genome is found in *Etheostoma uniporum* (Bossu & Near, 2009; Lang & Mayden, 2007; Ray et al., 2008).

Although previous studies of members of *Ceasia* have inferred regional patterns of genetic structure, a large-scale genetic study was needed to capture a broader picture of current and historical population genetic processes in this complex. Our study aims to apply riverscape genetics to investigate the *Ceasia* complex throughout a large portion of their North American range. Microsatellite genotypes of more than 380 fish collected from sites covering multiple watershed regions were used to address these questions: (1) What level of population structure and diversity exists between sampled *Ceasia* members across their range? (2) Do genetic clusters correspond to current species delineations? (3) Is there evidence of hybridization and introgression between *Ceasia* species? (4) Have landscape features (e.g., watershed boundaries) or historical events (e.g., glacial history) played a role in the underling genetic patterns?

2 | MATERIALS AND METHODS

2.1 | Specimen acquisition

Orangethroat darters were sampled between May 2016 and August 2017 using either a backpack electrofisher or a kick seine net along 50–100-m stream transects. Sampling locations were based on previous records gathered from museum collections, federal and state records, and local knowledge. Additional specimens were collected and donated to this study by other researchers and fish hobbyists. Specific site locations and details can be found in Table S1. The U.S. Geological Survey (USGS) uses a nested classification system of watershed designations to divide and subdivide the United States into successively smaller hydrologic units. We used the first three levels of that system to organize sampling sites: regional level (Hydrologic Unit Code [HUC] 2), then sub-region (HUC4), followed by the basin (HUC6) level. In all, 389 darters were collected from 52 sites across 20 river basins (HUC6) in six regions (HUC2) (Figure 1 and Table S1).

Specimens were identified to species in the field using colouration and body characteristics and confirmed in the laboratory using meristic characters following established keys (Kuehne & Barbour, 1983; Pflieger, 1997; Hubbs et al., 2004). Fish were euthanized in the field using tricaine methanesulfonate (MS-222) and preserved in 95% alcohol and later transferred to 70% alcohol for long-term storage. Collection sites were mapped against the most recent glacial maximums using data provided by the university of Koeln (https://crc806db.uni-koeln.de/). This study was carried out in strict accordance with the recommendations set forth by the University of Illinois at Chicago's Office of Animal Care and Institutional Biosafety (OACIB). This protocol was approved by the OACIB (protocol no.: 16-066).

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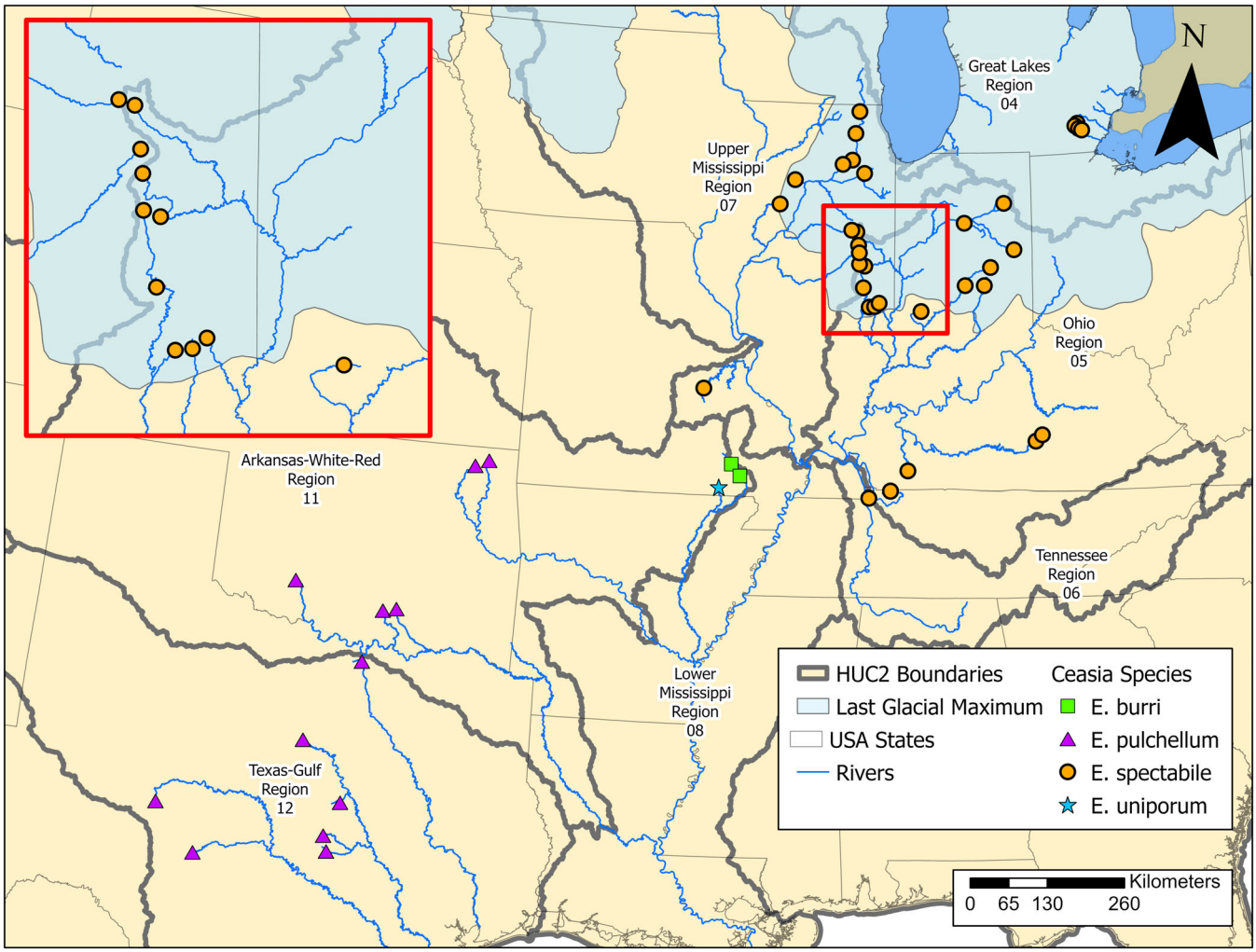

FIGURE 1 Approximate collection sites of orangethroat species complex specimens used in this study. Colors indicate species. Numbers and region names correspond to USGS HUC2 watershed designations. Inset shows regional boundary between Upper Mississippi River and Ohio River regions.

2.2 | Genetic analyses

Fin clips were taken from each fish ($n = 389$), and after DNA extraction using QIAGEN DNeasy Blood and Tissue Kit (cat. no.: 69504) or Lucigen QuickExtract kit (cat. no.: QE0905T), 384 individuals yielded usable genetic material. Ten microsatellite markers previously developed for rainbow (*E. caeruleum*) and orangethroat darters (*E. spectabile*) were used: Etsp203, Etsp216, Etsp227, Etsp240, Etsp401 (Hudman et al., 2008), Eca10EPA, Eca11EPA, Eca37EPA, Eca44EPA, and Eca46EPA (Tonnis, 2006). PCR was carried out in 10 μL reaction mixture containing approximately 40 ng of genomic DNA, 0.2 mM of each dNTP, 0.02 μM of forward primer, 0.2 μM reverse primer, 0.2 μM of fluorescently labeled M13 primer, 2.0 μL of 5× PCR Promega Buffer, 1.5 mM $MgCl_2$, and 0.1 μL of Promega GoTaq polymerase. An initial denaturing for 5 min at 95°C was followed by 35 cycles of 30s at 94°C, 30s at 55°C, and 30s at 72°C, followed by a final 5-min extension at 72°C. PCR fragments were detected on a 3730 Series Genetic Analyzer and individually scored using Applied Biosystems' GeneMapper v3.7 using an internal size standard modified from Maddox and Feldheim (2014).

Due to small sample sizes at several sites, individuals were aggregated at the HUC6 (basin) level for several genetic tests. For samples taken from multiple rivers within a basin, the closest downstream confluence was used for distance calculations. The number of individuals sampled in each basin along with the glaciation status of collection sites in each basin during the LGM is presented in Table 1.

Departures from Hardy–Weinberg equilibrium (HWE) were assessed with the pegas v1.1 R package (Kamvar et al., 2014) with a χ2 test using the false discovery rate (FDR) for each locus at the basin level. Linkage disequilibrium was based on the index of association proposed by Agapow and Burt (2001).

Descriptive genetic diversity indices were estimated for each locus and averaged across loci using GenAlEx v6.503 (Peakall & Smouse, 2006): number of alleles (N_a), effective number of alleles (A_e), and heterozygosity (observed – H_o and expected – H_e). Allelic richness (AR) was determined using the PopGenReport v3.0.4 R package

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TABLE 1 Watershed region, basin designations for aggregated sampling locations, and genetic diversity indices averaged over 10 loci.

Region	Basin	Glacial status	Species	N	N_a	A_e	H_o	H_e	AR	Pa
Great lakes										
	River Raisin	G	*ES*	20	4.3	2.371	0.462	0.475	3.033	1
Ohio River										
	Patoka-White	M	*ES*	43	14.10	7.090	0.768	0.485	6.242	7
	Wabash River	G	*ES*	112	14.60	5.796	0.630	0.804	5.728	34
	Green	U	*ES*	7	6.60	5.081	0.638	0.758	–	11
	Lower Cumberland	U	*ES*	3	2.50	2.186	-	0.550	–	4
Upper Mississippi										
	Lower Illinois	G	*ES*	52	9.70	4.450	0.508	0.587	4.616	17
	Upper Illinois	G	*ES*	26	5.40	2.571	0.413	0.485	3.25	7
	Upper Mississippi-Meramec	U	*ES*	1	2.00	2.00	1.000	0.500	–	2
Arkansas-White-Red										
	Neosho	U	*EP*	4	5.20	4.432	0.725	0.734	–	7
	Red-Lake Texoma	U	*EP*	4	2.4	1.970	0.600	0.428	–	0
	Red-Little	U	*EP*	5	5.40	4.521	0.720	0.709	–	16
	Upper White	U	*EB/EU*	27	11.30	5.755	0.659	0.768	5.703	40
	Verdigris	U	*EP*	11	8.60	5.553	0.727	0.760	5.778	10
	Washita	U	*EP*	10	4.80	3.418	0.608	0.593	3.898	2
Tennessee River										
	Lower Tennessee	U	*ES*	6	5.2	4.017	0.733	0.671	–	10
Texas Gulf										
	Little	U	*EP*	26	8.00	4.554	0.530	0.643	4.602	7
	Middle Brazos-bosque	U	*EP*	12	3.00	2.283	0.383	0.408	2.586	0
	Middle Colorado-Concho	U	*EP*	5	3.50	2.661	0.520	0.527	–	4
	Middle Colorado-Llano	U	*EP*	5	4.40	3.322	0.630	0.643	–	4
	Upper Trinity	U	*EP*	5	3.00	2.467	0.360	0.511	–	0

Note: Glaciated (G)—sites glaciated during LGM, unglaciated (U)—sites not glaciated during LGM, mixed (M)—both glaciated and unglaciated sample sites. Species: *ES*—*E. spectabile*, *EP*—*E. pulchellum*, *EB*—*E. burri*, *EU*—*E. uniporum*. Genetic diversity indices are averaged across loci: *N*—sample size, N_a—number of alleles, A_e—number of effective alleles, H_o—observed heterozygosity, H_e—expected heterozygosity, AR—allelic richness, Pa—private alleles. AR calculated for those basins with $N \geq 10$.

(Adamack & Gruber, 2014). Number of private alleles (Pa) and linkage disequilibrium were calculated using the poppr v2.9.3 R package (Kamvar et al., 2014). All indices were calculated at the global population level and the basin level for each locus with locus specific estimates for each index taken as a replicate observation and averaged for each basin. Differences between basins and the glacial history of basins for A_e, H_o, H_e, and AR were tested for significance using Kruskal-Wallis (stats v4.1.2 R package) with the Dunn's test showing pair-wise differences using the rstatix v0.7.0 R package (Kaassambara, 2023). A nested AMOVA was performed at the glacial history, HUC2, HUC6, and sampling site levels using the pegas v1.1 R package.

Genetic clusters were evaluated with STRUCTURE v2.3.4 using an admixture model with correlated allele frequencies (Pritchard et al., 2000). The number of possible clusters (K) tested was from $K = 1$ to $K = 15$ with 10 replicates for each K, a burn-in period of 100,000 followed by 600,000 replicates. Best K was identified by mean Delta K, ln P(K), MedMed K, MedMean K, MaxMed K, and MaxMean K using StructureSelector (Evanno et al., 2005; Li & Liu, 2018; Puechmaille, 2016). Once the best K was determined, individuals were assigned to their respective clusters and plotted using the pophelper v2.3.1 R package (Francis, 2017). Genetic clusters were also evaluated with the AIC method ("snapclust.choose.k" command) from the adegenet R package (Beugin et al., 2018; Jombart, 2008). The snapclust.choose.k command utilizes a maximum likelihood model-based technique and distance-based clustering combined with a goodness-of-fit statistic (AIC) to determine the optimal number of clusters (Beugin et al., 2018). Fish with less than 85% assignment to one cluster under both the STURUCTRE and AIC analyses were designated as "mixed ancestry" individuals. Genetic cluster assignments for individuals at each sample location were plotted using ESRI's ArcGIS Pro v2.9 software.

Principle component analysis (PCA) was calculated using the dudi.pca function of the ade4 v1.7–20 R package (Dray & Dufour, 2007). For the PCA, missing values were replaced by the mean allele frequency, and no scaling was used for the analysis. Jost's genetic

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distance (D_{Jost}) was calculated using the mmod v1.3.3 R package (Jost, 2008). The Nei (1987) F_{ST} values were calculated using the hierfstat v0.5–11 R package (Goudet, 2005). We tested both river distance and overland geographic distance for correlations with genetic distance. River distance is the length (km) of river required to move between two sites, calculated from current river maps using UTM zone 16 and the riverdist v0.15.4 R package. Current hydrogeographic connections were used to calculate distances between rivers that are not directly connected (e.g., along the shoreline of the Gulf of Mexico). An overland distance matrix for Euclidian distances (km) between points (shortest overland distance) was calculated using adegenet. The location used for distances between basins with multiple sample sites was the closest downstream confluence as described earlier. A Mantel test using the vegan v2.5–7 R package was used to test for correlations (isolation by distance [IBD]) between the genetic distances and geographic distances using the Spearman correlation distance with 999 permutations (Oksanen et al., 2005). The IBD significance was evaluated for the entire dataset and for four individual regions (Ohio River, Upper Mississippi, Arkansas-White-Red, and Texas-Gulf).

3 | RESULTS

A total of 384 fish were genotyped for this study; 270 identified as *E. spectabile*, 87 as *E. pulchellum*, 15 as *E. burri* Ceas and Page, and 12 as *E. uniporum* Distler. There may have been additional species represented among the *E. spectabile* collected from the Green, Lower Cumberland, and Lower Tennessee as the collection locations overlap the ranges of *Etheostoma tecumsehi* Ceas and Page and *Etheostoma lawrencei* Ceas and Burr (Green River), Mamequit darter (Lower Cumberland), and *Etheostoma bison* Ceas and Page (Lower Tennessee). However, we were not confident in our identification as there were few specimens of each, and colouration and morphological characters did not clearly resemble diagnostic characters (Ceas, 1997; Ceas & Burr, 2002; Ceas & Page, 1997). These questionable individuals were grouped as *E. spectabile* per Bossu and Near (2009).

3.1 | Genetic diversity

All fish were genotyped for the 10 microsatellite loci, with 3.5% missing data overall. The number of alleles per locus ranged between 20 (Etsp240) and 84 (Eca11EPA) and expected heterozygosity ranged from 0.850 (Etsp240) to 0.943 (Eca46EPA) (Table 1). There was significant linkage disequilibrium for the global population (Index of association [Ia] = 1.083, $\bar{r}d = 0.126$, $p = 0.001$), with strongest associations between Etsp240 and Eca10EPA ($\bar{r}d = 0.422$) and Etsp401 and Eca37EPA ($\bar{r}d = 0.314$). However, linkage disequilibrium accounted for only *c.* 12% of the variation, so we did not discard any loci.

Deviations from HWE were tested for each locus for each basin, except for the Lower Cumberland basin where all samples had missing data for locus Eca44EPA (Table S2) as it failed to amplify. Overall, 14.6% of the combinations of loci and basins deviated from HW expectations when the FDR correction was applied, with Etsp401 accounting for 25% of significant deviations (Table S2).

We assigned basins to glaciated and unglaciated designations depending on glacial history of their sampling sites to test for difference in genetic diversity. One basin (Patoka-White) had both glaciated and unglaciated collection sites, so it was designated as mixed. The differences between the three classifications in H_o ($\chi^2 = 9.355$, $df = 2$, $p = 0.009$) and AR ($\chi^2 = 10.065$, $df = 2$, $p = 0.007$) were significant using Kruskal-Wallis. The Dunn's test showed that fish populations from glaciated sites had significantly lower H_o and AR than fish populations from unglaciated sites. There was no difference between the mixed basin and either the unglaciated or glaciated basins.

The results of the AMOVA indicated that most of the variation (*c.* 54%) is attributed to an unknown factor (Table 2). The most variation (*c.* 22%) can be attributed to differences between sites followed closely by differences between regions. The least variation was partitioned between the glaciated and unglaciated areas.

3.2 | Genetic clustering

The STRUCTURE analysis found K = 3 and K = 8 genetic clusters as determined by the ΔK method (Figure 2). For MedMed K, MedMean K, MaxMed K, Max Mean K, the best K was the maximum tested (15), and for ln P(k) it was 11 (Figure S1). These high K values were not biologically meaningful and were not explored further. For the 384 genotyped individuals under K = 3 classification, 78 were assigned to Cluster 1 (blue in Figure 2a), 102 to Cluster 2 (red), and 128 to Cluster 3 (green) using the 85% classification. The remaining 76 fish were of mixed ancestry and were assigned to mixed clusters. There were 56 fish that had a high proportion of Clusters 1, 2, and 3 (Figure 2a) that were designated as purple (Figure 3). These fish were *E. spectabile* collected from the Embarras River subbasin of the Wabash and *E. burri* from the Upper White basin. There were

TABLE 2 AMOVA of populations of *Etheostoma spectabile*.

Source of variation	*df*	Sum of squares deviations	Percentage of variation	phi
Glacial Group	1	90.71	2.94	0.129
HUC 2	6	278.62	14.88	0.651
HUC 6	13	187.75	6.54	0.286
Site	29	280.60	22.00	0.962
Error	334	783.73	53.64	2.346
Total	383	1621.40		

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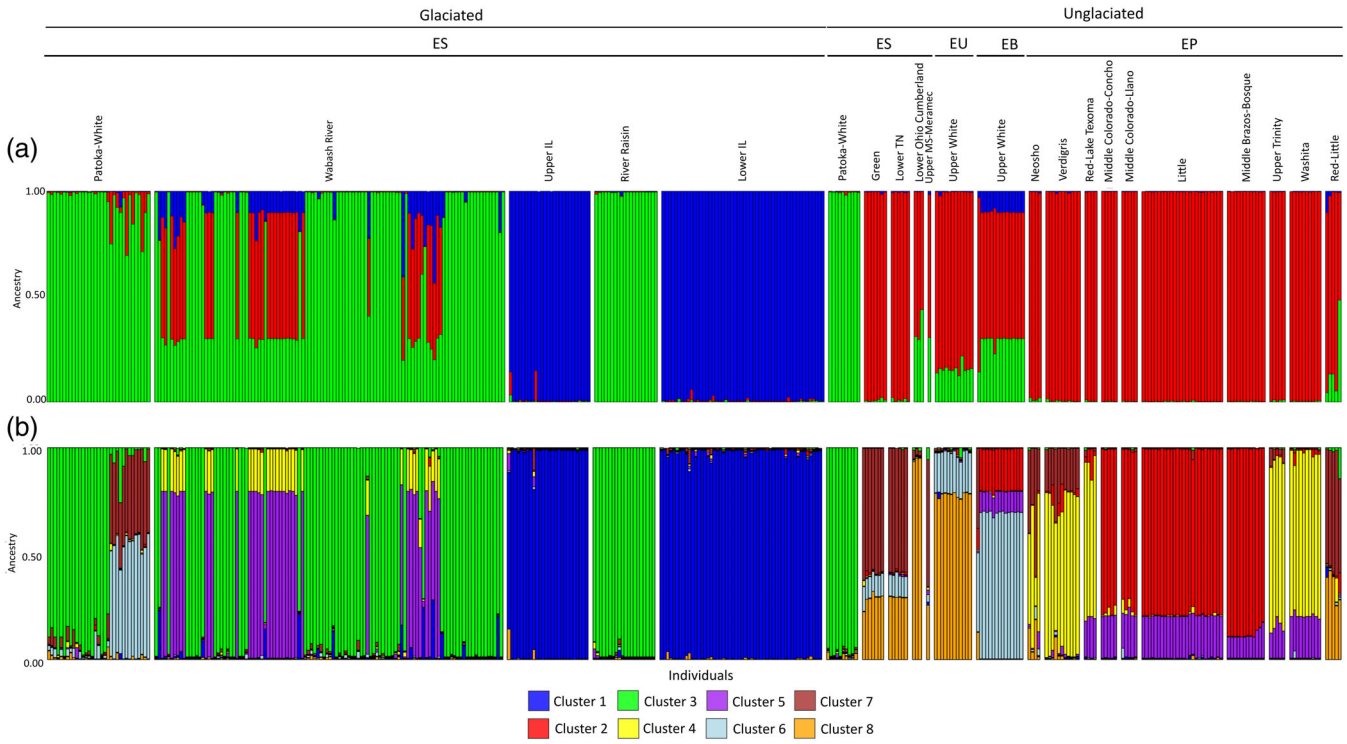

FIGURE 2 Results from STRUCTURE for (a) K = 3 and (b) K = 8. Each color corresponds to a distinct genetic cluster, and each bar represents an individual's assigned ancestry to each cluster. Names above chart correspond to river basin (HUC6). EB—*E. burri*, EP—*E. pulchellum*, ES—*E. spectabile*, and EU—*E. uniporum*. Glaciated and unglaciated correspond to areas that were/were not covered in ice during the last glacial maximum.

16 individuals that had high proportions of Clusters 2 and 3 that were designated as an orange cluster (Figure 3). This mixed cluster consisted of *E. spectabile* individuals from Lower Cumberland, Upper Mississippi-Meramec, and the Patoka-White basins, and *E. uniporum* from the Upper White basin. Lastly, four *E. spectabile* that showed a high proportion of Clusters 1 and 3 and were designated as a yellow cluster all found at one location in the Wabash basin (Figure 3). Fish assigned to the red cluster were all collected at sites that were not glaciated during the LGM, c. 26.5 -19kya (Figure 3), whereas fish assigned to the blue and green clusters were predominantly from sites that had been glaciated (Clark et al., 2009).

Cluster assignments under the STRUCTURE K = 8 scenario are shown in Figure 2b and mapped in Figure S3. Due to many individuals showing admixture, a simple majority rule was used for cluster assignments for K = 8. The cluster shown in green (Wabash Great Lakes Cluster) was the largest cluster with 123 individuals. The blue cluster contained 78 fish from the Illinois River, and the red cluster contained all 48 individuals from the Texas-Gulf region. Fish from the purple cluster ($N = 39$) were from the Embarras River. The cluster shown in yellow contained 33 fish from the Arkansas-Red River basin. All the individuals identified as *E. burri* (brook darter) and orangethroats from the Patoka-White basin belonged to the cluster shown in teal (Brook Darter Cluster, 26 individuals). Cluster 2 (brown, Highlands Plateau Cluster) had 22 individuals from four different regions. Individuals identified as *E. uniporum* and three orangethroats from the Cumberland basin belonged to Cluster 8 (orange, Current Darter Cluster, 15 individuals). Several basins showed large numbers of individuals that had admixed ancestry. Overall, individuals assigned to the Illinois, Wabash Great Lakes, and Embarras clusters were predominantly found in previously glaciated areas and the Highlands Plateau, Arkansas-Red River, Current Darter, and Gulf clusters were in unglaciated areas. The Brook Darter Cluster was found in glaciated and unglaciated areas.

The AIC cluster analyses returned an optimal K = 5 (Figure S2) with the assigned ancestry plot shown in Figure 4. Predominantly glaciated and unglaciated clusters were still identified, but there was an additional structure within each of these groups, with AIC analysis delineating three clusters in the glaciated and two in the unglaciated areas (Figure 5). Table S5 shows the number of individuals from each basin assigned to each cluster and the number of individuals that did not meet the 85% assignment criteria (mixed). Overall, most individuals within a basin were assigned to one genetic cluster; however, most clusters included individuals from multiple basins. Cluster 3 (green) had 124 individuals from the Wabash, River Raisin, and Patoka-White basins (hereafter "Wabash Great Lakes Cluster"). Cluster 4 (purple) was only comprised of 39 individuals from the Embarras River, a subbasin of the Wabash basin (hereafter "Embarras Cluster"). Cluster 5 (blue) had 78 individuals coming from the Upper and Lower

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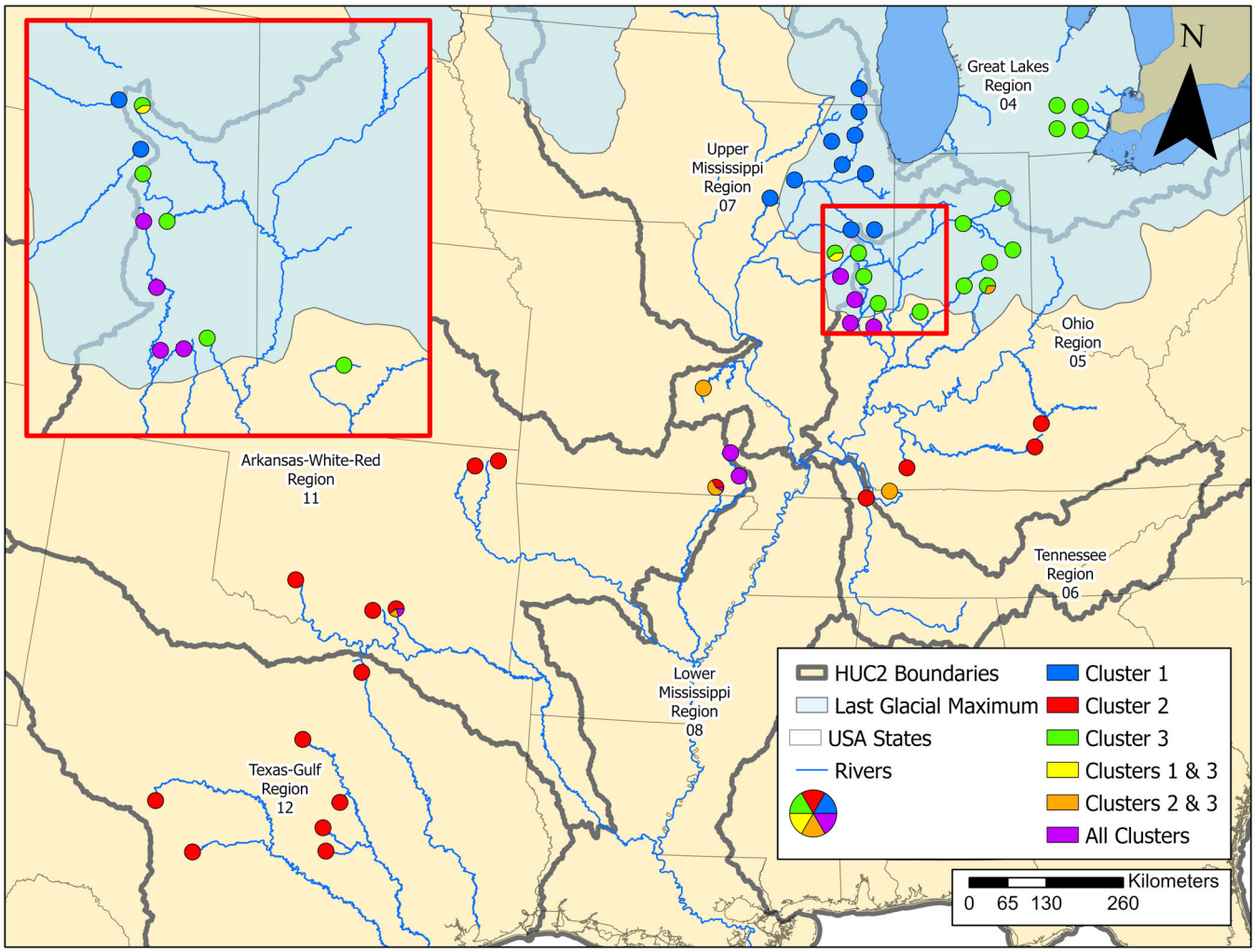

FIGURE 3 Orangethroat darter clade collection sites with colored circles corresponding to the proportion of individuals at the site that had predominant assignment to each genetic cluster identified in the STRUCTURE analysis for K=3. Inset shows region boundary between Upper Mississippi River and Ohio River regions.

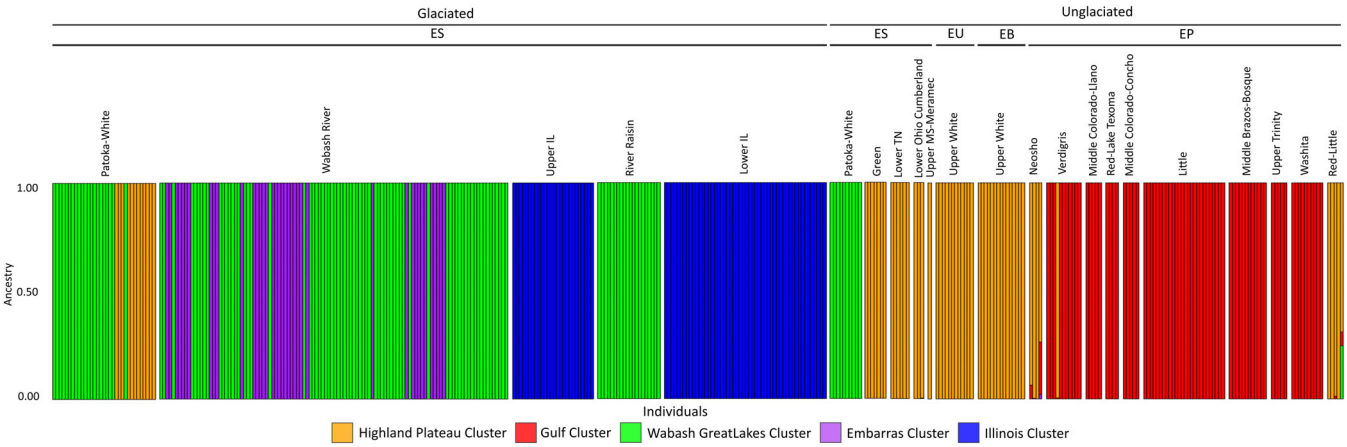

FIGURE 4 Assigned ancestry plot using AIC criteria determining optimal K of 5. Each color corresponds to a distinct genetic cluster, and each bar represents an individual's assigned ancestry to each cluster. Names above chart correspond to river basin (HUC6). EB—*E. burri*, EP—*E. pulchellum*, ES—*E. spectabile*, and EU—*E. uniporum*. Glaciated and unglaciated correspond to areas that were/were not covered in ice during the last glacial maximum.

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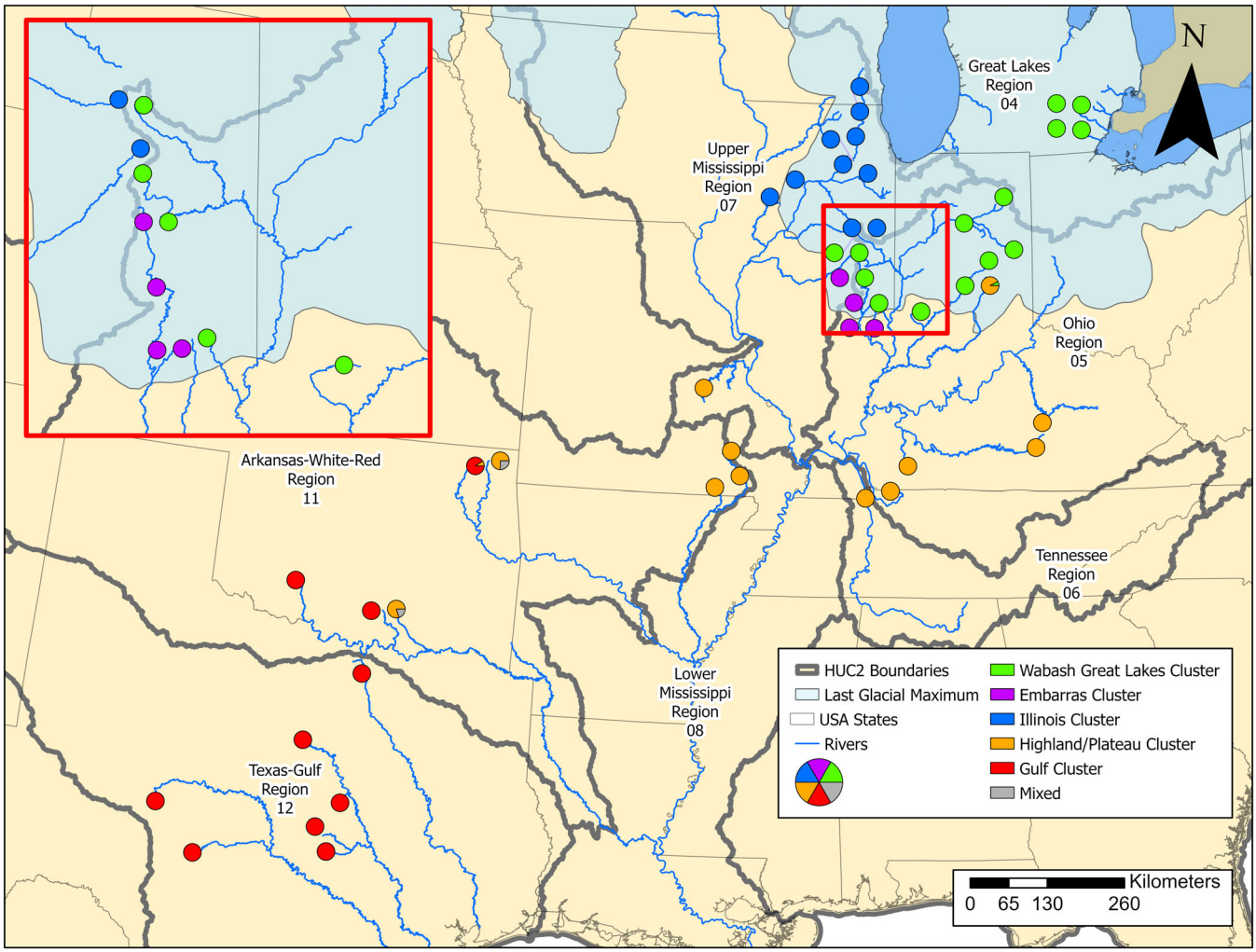

FIGURE 5 Cluster assignments at sampling locations under the AIC method and a K=5. Pie graphs represent proportion of individuals at the site assigned to a given cluster. Individuals with less than 85% assigned ancestry in either cluster are shown in gray.

Illinois basins (hereafter "Illinois Cluster"). Cluster 1 (orange) had 64 individuals from the most basins (9) (hereafter "Highlands Plateau Cluster"). Finally, Cluster 2 (red) was the only cluster found in the Gulf Region (hereafter "Gulf Cluster").

The results of the PCA (Figure 6) indicate that samples from the Upper and Lower Illinois River basins are clearly distinct from those of other basins. Wabash River basin fish overlap with those of the Patoka-White and River Raisin basins and are mostly separated from the other basins. However, Wabash River fish do also overlap with *E. spectabile* sites from the Highland Plateau cluster of the AIC analysis (e.g., Lower Cumberland, Lower Tennessee, Upper Mississippi-Meramec) including the Upper White basin where *E. burri* and *E. uniporum* were collected. The Texas-Gulf and several AK-White-Red region basins, where *E. pulchellum* was collected, are clustered at the top of the figure, showing little or no overlap with basins outside of these regions. A second PCA was conducted to compare glaciated and unglaciated basins (Figure 7). This analysis showed that there is separation of the glaciated and unglaciated areas.

3.3 | Isolation by distance

For IBD, both the overland distance (Figure 8a) and river distance (Figure 8b) were significantly correlated with F_{ST} distance (river: Mantel Statistic [r] = 0.406, $p = 0.0.004$, land: $r = 0.367$, $p = 0.001$). Both the river and overland distance were also significantly correlated with D_{Jost} (river: $r = 0.270$, $p = 0.001$ L; land: $r = 0.531$, $p = 0.001$). Within regions, no significant IBD was present for either overland distance (Figure 8c) or river distance (Figure 8d) for either F_{ST} or D_{Jost}.

4 | DISCUSSION

There are 50 families with over 1300 native fish species that occupy North American rivers (Warren et al., 2020). The drainage patterns, flow regimes, and hydrology of these waterbodies have been impacted by geological events, most notably the glacial cycles of the Pleistocene. Advancing and retreating glaciers altered the river morphology and flow patterns, and as a result, fish moved into new

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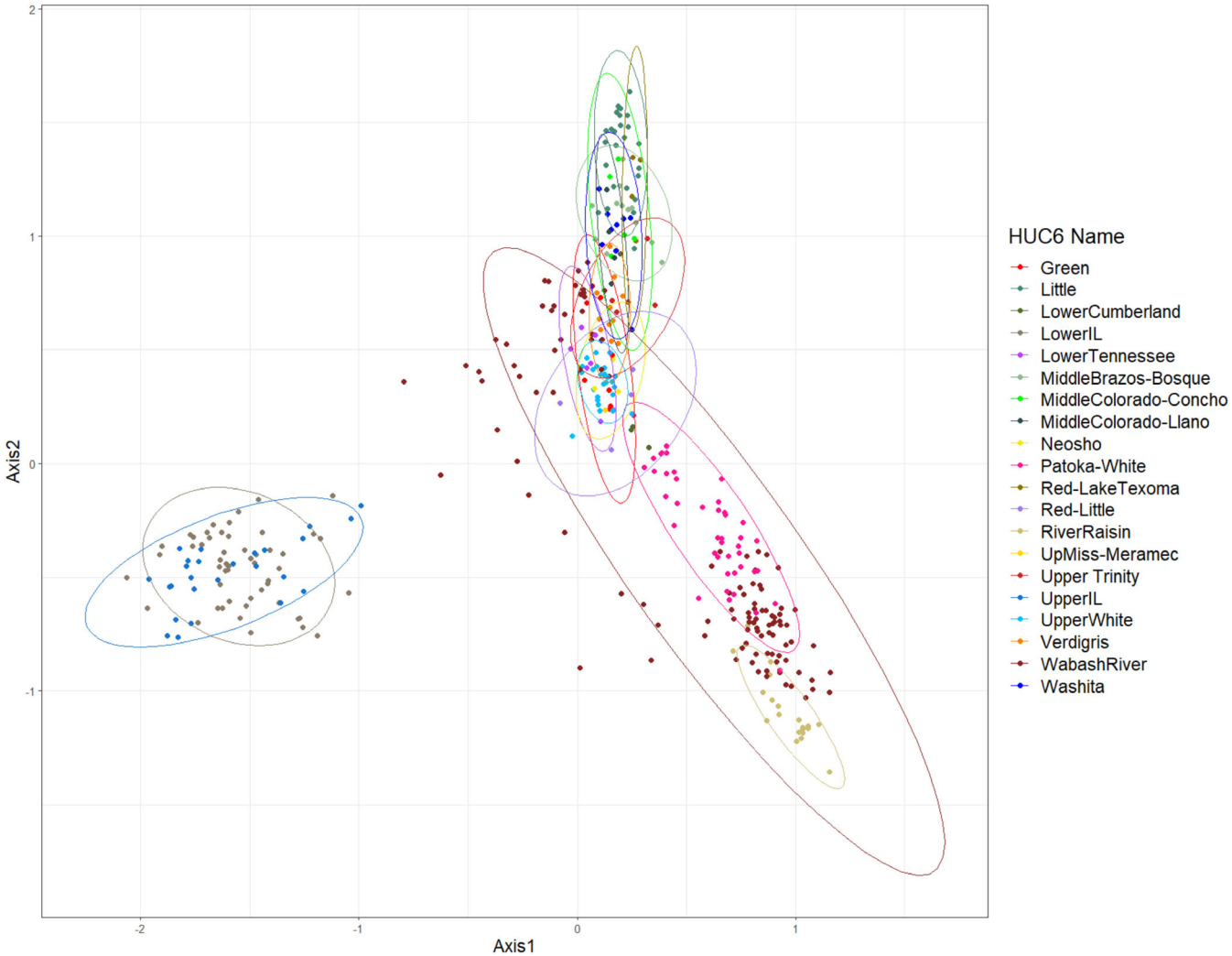

FIGURE 6 Principle component analysis (PCA) of *Ceasia*. Name indicates the basin where samples were collected.

watersheds, were extirpated from watersheds where they previously occurred, or were isolated in newly formed watersheds. These dynamics impacted many freshwater species, including members of *Ceasia*. This complex includes *E. spectabile* which has an expansive range that includes regions that were covered by glaciers as recently as 20,000 years ago as well as regions that were never glaciated (Ceas & Page, 1997; Page & Burr, 2011). Another species, *E. pulchellum*, has a large southern range not directly impacted by glaciations. Other members of *Ceasia* have very restricted ranges in rivers of the Ozark and Eastern highland areas. Thus, the orangethroat complex provides an excellent opportunity to explore factors shaping genetic structure and diversity in Central North America riverine systems, an important region of freshwater fish diversity.

Our sampling covered much of the range of *Ceasia* and included at least four recognized species in the complex. Analyses of microsatellite genotypes identified strong but surprising patterns of genetic structure. One consistent finding was a pattern of differentiation that corresponds roughly to the boundary of the LGM. The genetic clusters delineated by the LGM are most strongly evident in the results of the structure analysis under the K = 3 scenario, with most individuals belonging to either two genetic clusters largely confined to the previously glaciated regions or a cluster found throughout the unglaciated parts of their range (Figures 2a and 3). The high level of differentiation of the two clusters found in the glaciated portion of the *E. spectabile* species range was somewhat surprising but was consistent through all our analyses. Notably, fish from the Upper and Lower Illinois River basins comprised an extremely distinct genetic cluster, as shown in the PCA (Figure 6), which suggests that there has been little gene flow into the Illinois River from directly adjacent drainages. This also suggests that the Illinois River population originated from its own refugium, something previously suggested by Bossu et al. (2013) based on nuclear and mitochondrial gene phylogenies. The refugia is likely somewhere in the lower reaches of the Illinois River or some place in the Ozarks not sampled for this study. A similar Ozark to upper Mississippi drainage dispersal was proposed for the northern hogsucker (*Hypentelium nigricans*, Lesueur) by Berendzen et al. (2003).

The second *E. spectabile* cluster in the previously glaciated portion of the range was comprised of fish from the Wabash River, River

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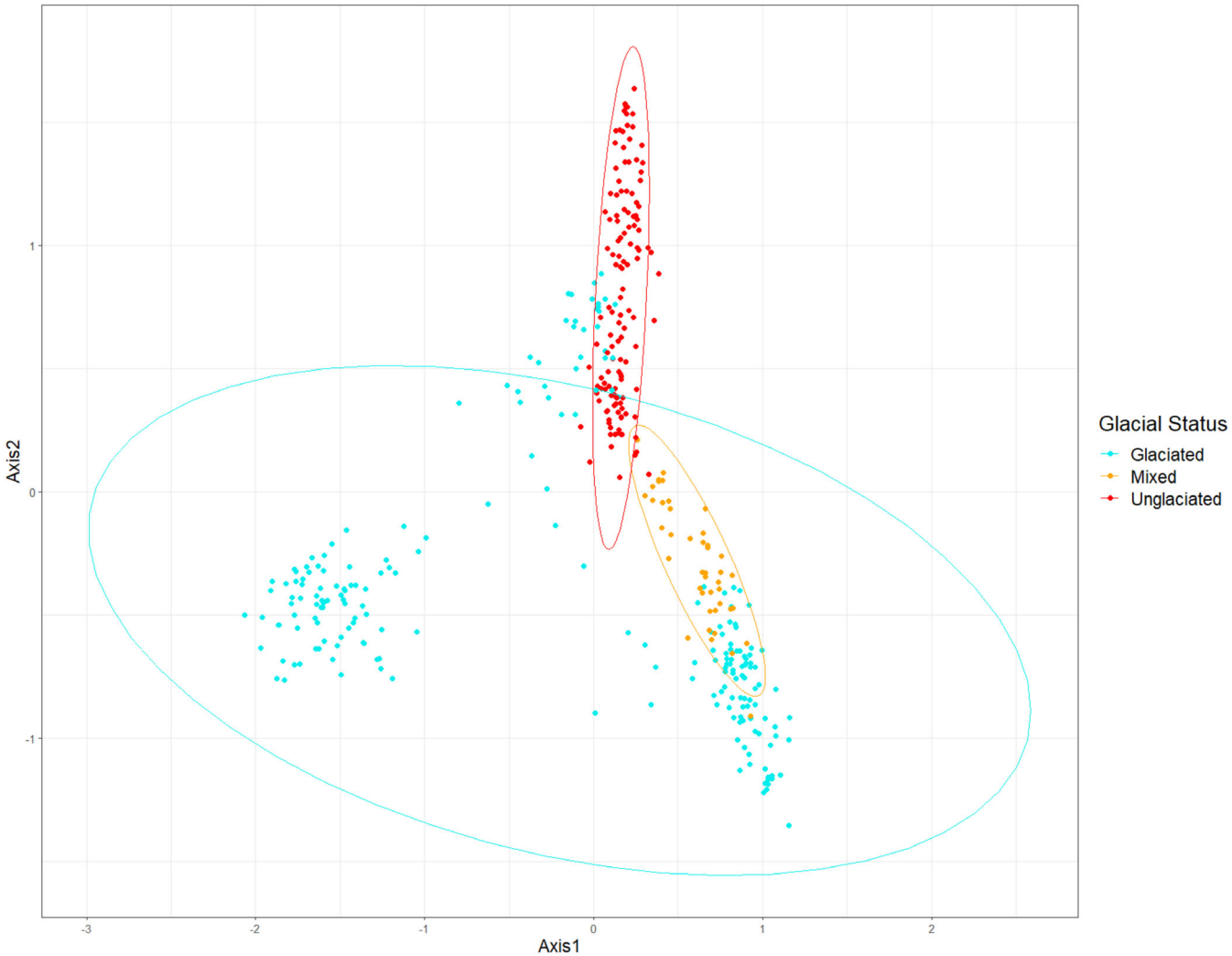

FIGURE 7 Principle component analysis (PCA) of *Ceasia* showing distribution of samples taken from glaciated, unglaciated, or the one basin that had both glaciated and unglaciated sampling sites ("Mix").

Raisin, and Patoka White basins ("Wabash Great Lakes Cluster"). Like the Illinois River Cluster, this cluster of fish was found to be genetically quite distinct in various analyses (e.g., Figures 6 and 7), but in contrast to the Illinois Cluster, this cluster includes fish from drainages that are no longer directly connected. The distribution is likely explained by temporary connections between basins starting around 14,000 years bp when the Wabash River flowed north into Lake Maumee, the predecessor to Lake Erie (Bailey & Smith, 1981). Thus, the Wabash River may have acted as a source population for the *E. spectabile* now in the Lake Erie basins and was itself likely derived from a glacial refugium separate from the Illinois River, as previously proposed (Bossu et al., 2013). This finding may be particularly important for conservation efforts for *E. spectabile* in Michigan. The species is only found in three watersheds (including the River Raisin) in southeast Michigan and is listed as a critically imperiled species of special concern in the state (https://mnfi.anr.msu.edu/). Any restocking or reintroduction programme should consider the close genetic affinities with fish in the Wabash Great Lakes Cluster. The refugia that served as the source for the Wabash Great Lakes Cluster is likely to the south or east of the Wabash River and a product of post-glacial expansion when portions of the Wabash River were connected to other drainages (e.g., the Teays River). This southern/southeastern refugia was also posited as a second refugial population of the northern hogsucker (Berendzen et al., 2003).

Although the structure analysis shows genetic clusters predominantly delineated by the LGM, some individuals showed higher levels of mixed ancestry, belonging to two or more clusters. The Embarras River subbasin of the Wabash basin was a consistent location where individuals either showed mixed ancestry (Figure 2) or belonged to a distinct cluster (Figure 4). These fish are likely derived from fish that dispersed from a third refugia in the Eastern or Ozark highlands and/or is a basin with a high level of gene flow from the Wabash and Illinois basins and a southern, unglaciated location. It was previously postulated that *Ceasia* from the Embarras River likely came from a separate source than the rest of the Wabash River, either from a stream capture or flooding event in the Embarras River headwaters

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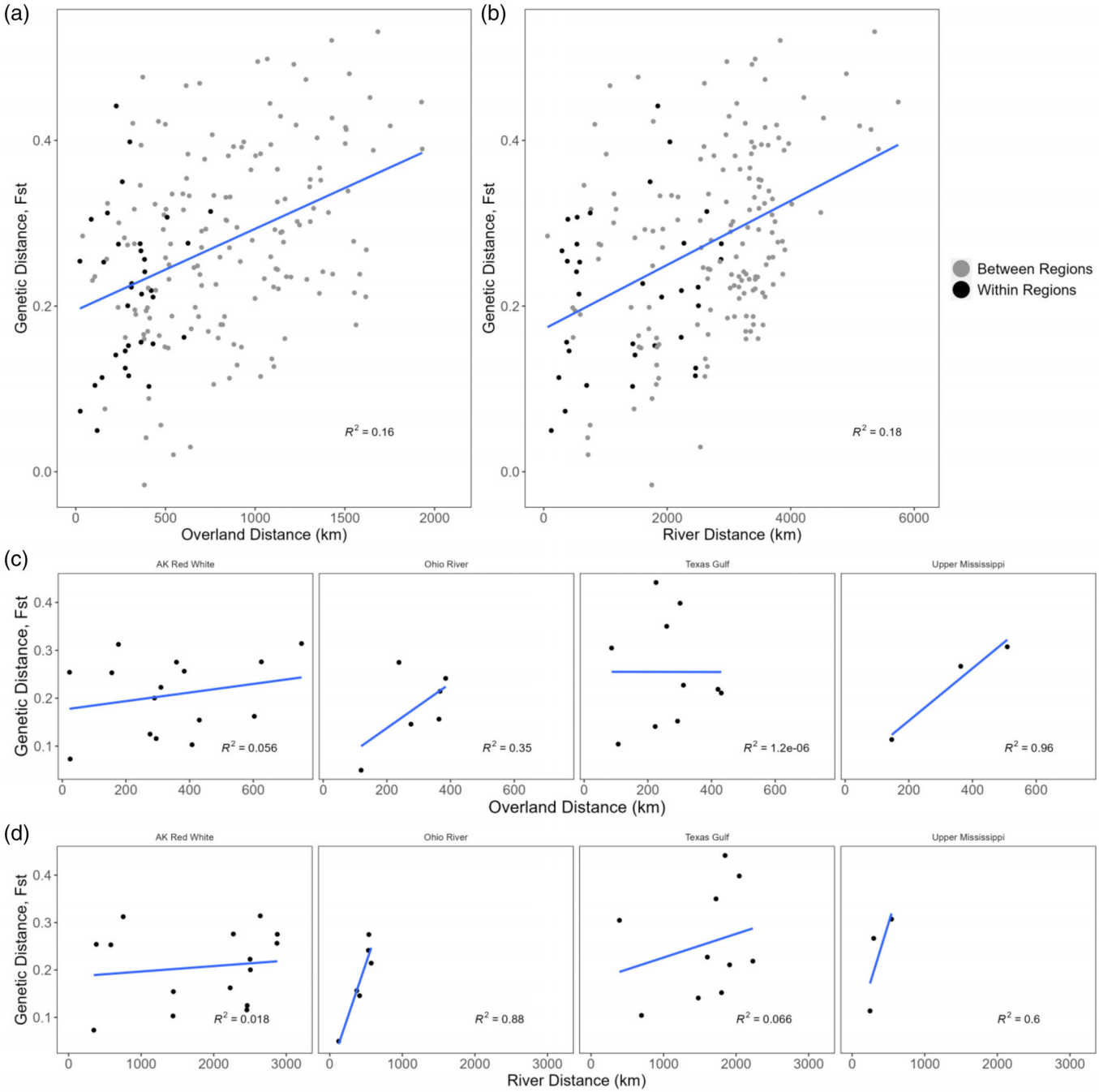

FIGURE 8 F_{ST} isolation by distance plots. Segregation of allele frequencies is shown between pairs of basins *v.* geographic (river/land) distance. (a, b). All basin pairs in this study with comparisons within regions shown in black and between regions in gray for overland distance (a) and river distance (b). (c, d). Within region site comparisons, grouped by regions for overland distance (c) and river distance (d). The Tennessee River and Great Lakes regions were omitted because one basin was sampled in each region.

or post-glacial colonization from the Osage River (Bossu et al., 2013). Our results support the separate refugia hypothesis.

Fish from previously glaciated basins showed strong genetic differentiation, but they had somewhat lower genetic diversity than fish from unglaciated basins, with significantly lower values of H_o and AR (Table 1). Post-glacial expansion might result in reduced genetic variation either because of population bottlenecks during recolonization or genetic drift in small refugial populations prior to expansion. Previous smaller studies of the ecologically similar rainbow darters (*E. caeruleum*) as well as other fishes have shown that species with ranges that span glaciated and unglaciated areas have lower genetic diversity in glaciated areas compared to unglaciated areas (Bernatchez & Dodson, 1991; Billington & Hebert, 1991; Haponski & Stepien, 2008; Hewitt, 1996, 2000; Luiken et al., 2021; Sepulveda-Villet et al., 2009; Sepulveda-Villet & Stepien, 2012).

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Fish from the unglaciated region of *Ceasia* included at least four recognized species (*E. spectabile*, *E. burri*, *E. uniporum*, and *E. pulchellum*). Under the K = 3 scenario, a single cluster dominated this entire unglaciated area. The PCA (Figures 6 and 7) also reveals little differentiation among these populations, especially compared to the two clusters from the glaciated regions. STRUCTURE's K = 8 scenario and the AIC analysis (K = 5) identifies some substructure in the unglaciated regions (Figure 5 and Figure S3). In the AIC analysis, most of the fish coming from the Texas-Gulf Region and the west portion of the Arkansas-White-Red Region grouped together ("Gulf Cluster"; Figure 5). Under the K = 8 structure scenario, these two regions each had a unique cluster ("Gulf Cluster" and "AR Red River Cluster"; Figure S3) not found elsewhere. The PCA analysis shows there is some genetic distinctness for fish from these regions (Figure 6) as they mostly do not overlap with other basins or the other three species. These fish were identified as *E. pulchellum*, a species found in several rivers that are currently unconnected. Their current distribution is likely a result of ancient river connections facilitated by the Ancient Plains Stream that ran through these now-separate basins. Previous studies supported this scenario and suggested that *E. pulchellum* derived from a refugium of *E. spectabile* in the Ozark Highlands that was isolated during the pre-Illinoian glaciation event (Bossu et al., 2013; Distler, 1968; Wiseman et al., 1978).

The AIC analysis identified a "Highland/Plateau Cluster" that contained fish collected from the Ozark Highlands, Eastern Highlands, and Oklahoma Plains. This cluster contains all four species collected for this study. Under the K = 8 STRUCTURE analysis, the Highland/Plateau Cluster is comprised of some *E. spectabile* and *E. pulchellum* individuals, with *E. burri* and *E. uniporum* comprising distinct clusters (Brook and Current clusters, respectively). Both *E. burri* and *E. uniporum* have allopatric ranges limited to drainages within the Ozark Highlands (Upper White River basin) and are thought to be recently diverged species isolated by high levels of behavioral isolation but low levels of genetic differentiation (Moran et al., 2017). Several *E. spectabile* individuals outside of the *E. burri* and *E. uniporum* ranges were also included in the Brook and Current clusters (Figure S3). These *E. spectabile* individuals were collected at two sites where *E. caeruleum* were also collected. Avoidance of hybridization with *E. caeruleum* has been shown to drive reproductive character displacement and agonistic character displacement in *Ceasia*, which incidentally contributes to allopatric divergence within the species complex (Moran et al., 2017; Moran & Fuller, 2018a, 2018b).

Why would fish from glaciated regions show more genetic structure than fish from unglaciated regions? Several factors might be involved. First, fish occupying the formerly glaciated regions may not have had time to diversify significantly from their refugial populations, resulting in well-defined genetic clusters that conform tightly to basin boundaries. Likewise, there has been less time for stream capture events that can play a role in moving fish between unconnected but adjacent locations. Hybridization and introgression may also explain the lack of genetic structure and species delineation in the unglaciated parts of the range, especially in the regions of high diversity and endemism. Extensive introgression and hybridization have been reported throughout *Etheostoma* and more narrowly between *Ceasia* members and other species (Bossu & Near, 2009, 2013; Lang & Mayden, 2007; Ray et al., 2008). Hybridization was suggested to be the main driver for the evolution of and diversity seen in the orange-throat complex in an early study (Distler, 1968). More recently, studies have shown that some of the diversity seen in *Ceasia* is driven by sympatric male–male interactions particularly in areas of sympatry with *E. caeruleum* (Moran et al., 2018; Moran & Fuller, 2018a, 2018b). Movement of microsatellite alleles and homogenization of allele frequencies through hybridization may obscure genetic differentiation despite morphological or behavioral differences between species. Furthermore, these species could be recently diverged and would only be distinguished by markers that covered more of the genome or were linked to male mating behavior.

AUTHOR CONTRIBUTIONS

John T. Belcik and Mary V. Ashley conceived the study. John T. Belcik collected data, performed the analyses, and led the writing with input from Mary V. Ashley.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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